# Report

**Team members:**

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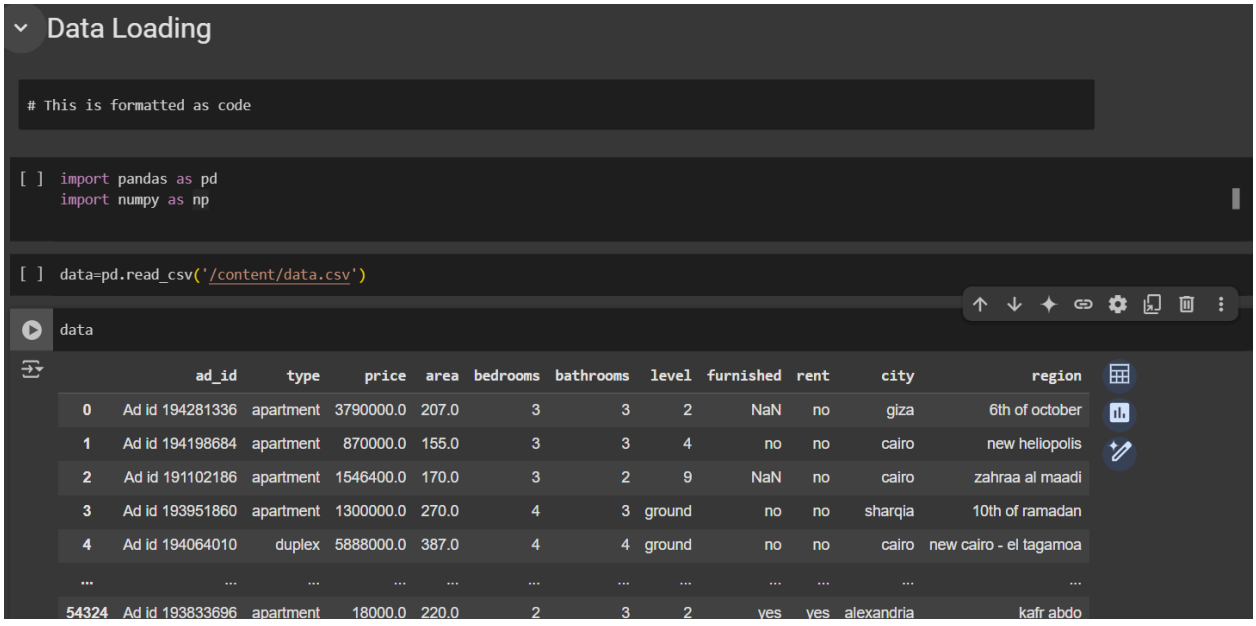
# Step 1: Data Loading

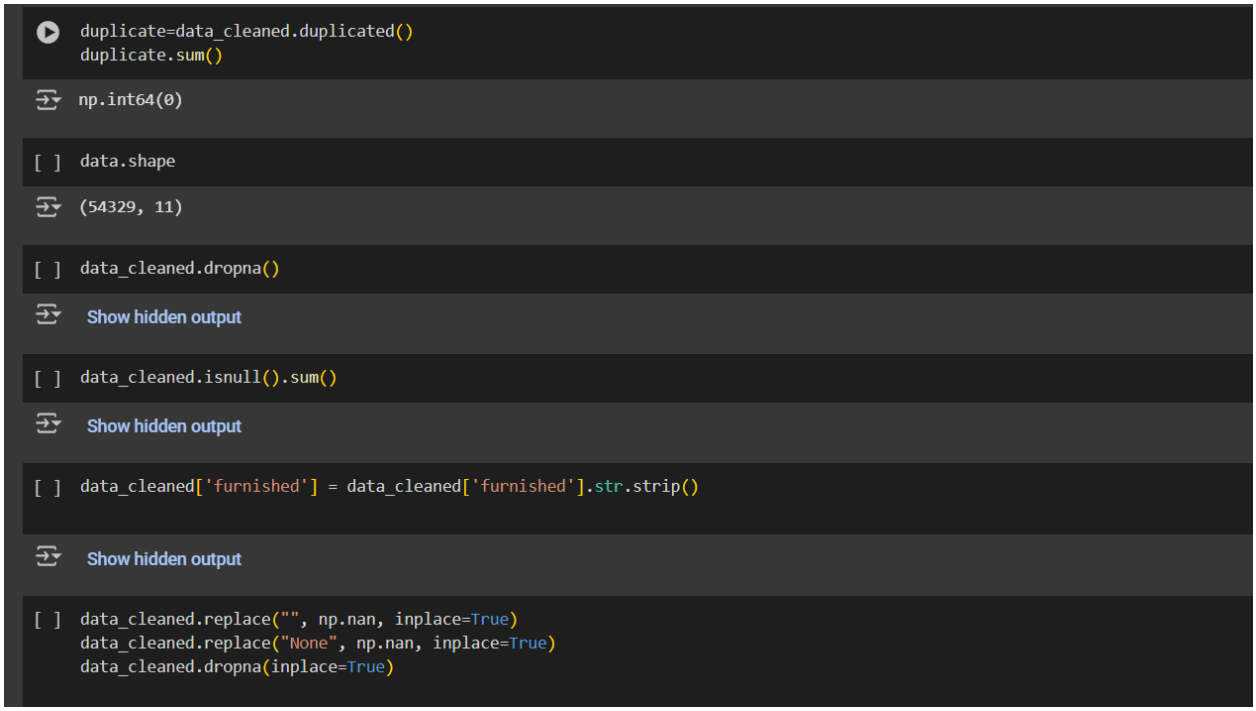
Loading data from CSV File




| | ad_id | type | price | area | bedrooms | bathrooms | level | furnished | rent | city | region |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Ad id 194281336 | apartment | 3790000.0 | 207.0 | 3 | 3 | 2 | NaN | no | giza | 6th of october |
| 1 | Ad id 194198684 | apartment | 870000.0 | 155.0 | 3 | 3 | 4 | no | no | cairo | new heliopolis |
| 2 | Ad id 191102186 | apartment | 1546400.0 | 170.0 | 3 | 2 | 9 | NaN | no | cairo | zahraa al maadi |
| 3 | Ad id 193951860 | apartment | 1300000.0 | 270.0 | 4 | 3 | ground | no | no | sharqia | 10th of ramadan |
| 4 | Ad id 194064010 | duplex | 5888000.0 | 387.0 | 4 | 4 | ground | no | no | cairo | new cairo - el tagamoa |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 54324 | Ad id 193833696 | apartment | 18000.0 | 220.0 | 2 | 3 | 2 | yes | yes | alexandria | kafr abdo |

# Step 2: Data Cleaning

Handling Missing & Invalid Entries:


**Other actions:**

- All blank strings (`""`) and `"None"` entries were replaced with `np.nan`.
- Final removal of `NaN` values from critical column

# Outlier Handling

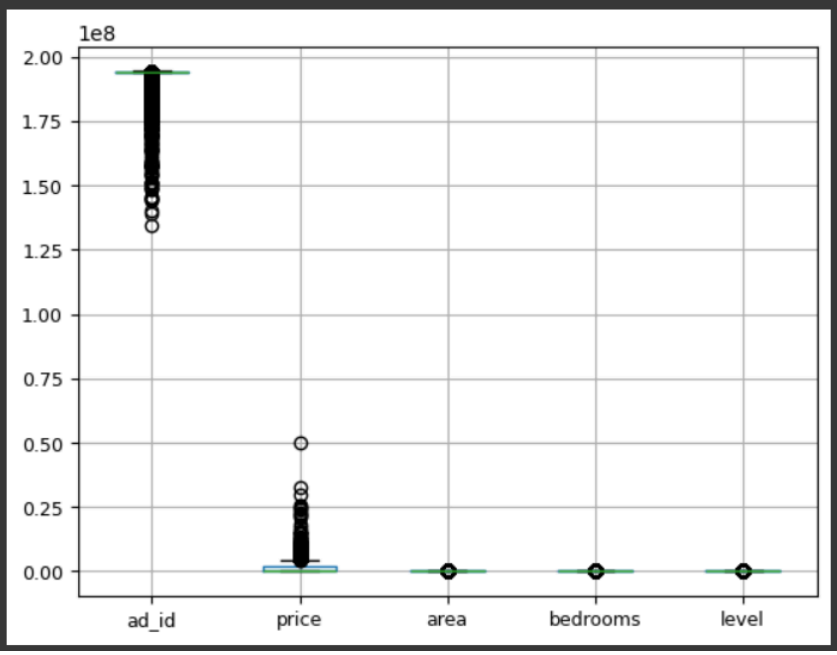
- Used **boxplots** to visualize numeric columns.

- Applied **clipping** to the `price` column (5th–95th percentile)
- Re-visualized boxplots after clipping.

**Before Check Outlier:**

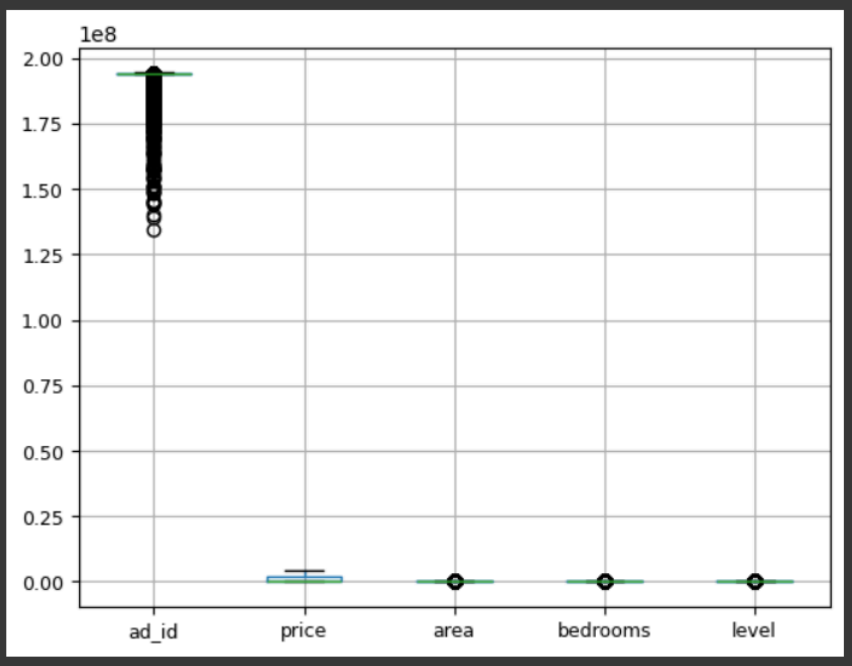
```
[ ]  numerical_columns=['ad_id','price','area','bedrooms','level']

     boxplot =data_cleaned.boxplot(column=numerical_columns, grid=True, fontsize=9)
     plt.show()
```




**After Remove Outliers:**

```
[ ]  boxplot =data_cleaned.boxplot(column=numerical_columns, grid=True, fontsize=9)
     plt.show()
```




## . Feature Transformation

- Converted appropriate columns to `int` or `float`.
- Standardized all column names by stripping spaces.
- Ensured `furnished`, `rent`, and `type` are within allowed categorical values.



- Handling the categorical  values from the numerical columns by replacing it with numbers.

# Step 3 : Data Validation Using Pandera

Defined and applied a `DataFrameSchema` using `pandora` to ensure

```
!pip install pandera
import pandera as pa
import pandas as pd
from pandera import Column, DataFrameSchema, Check, Index
```

Show hidden output

```
import pandera as pa

schema = pa.DataFrameSchema({
    "price": pa.Column(float, pa.Check(lambda x: x >= 0)),  # Price must be non-negative
    "area": pa.Column(float, pa.Check.in_range(1, 1000)),  # Area range
    "bathrooms": pa.Column(int, pa.Check.in_range(1, 10)),  # Bathrooms between 1 and 10
    "bedrooms": pa.Column(int, pa.Check(lambda x: x > 0)),  # Bedrooms must be positive
    "ad_id": pa.Column(int, pa.Check(lambda x: x > 0)),  # Ad ID must be valid
    "type": pa.Column(str, pa.Check.isin(["apartment", "duplex", "penthouse", "studio", "room"])),  # Property types
    "level": pa.Column(float, pa.Check(lambda x: x >= 0)),  # Level must be non-negative
    "furnished": pa.Column(str, pa.Check.isin(["yes", "no"])),  # Only 'yes' or 'no' allowed
    "rent": pa.Column(str, pa.Check.isin(["yes", "no"])),  # Only 'yes' or 'no' allowed
    "city": pa.Column(str),  # Check if city names exist
    "region": pa.Column(str),  # Ensure region is valid
   "level": pa.Column(int, pa.Check(lambda x: x >= 0))
})
```

```
validated_data = schema.validate(data_cleaned)
print(validated_data.head())
```

```
validated_data = schema.validate(data_cleaned)
print(validated_data.head())
```

```
      ad_id       type      price   area  bedrooms  bathrooms  level  \
1  194198684  apartment   870000.0  155.0         3          3      4
3  193951860  apartment  1300000.0  270.0         4          3      0
4  194064010     duplex  5888000.0  387.0         4          4      0
5  194021448  apartment   950000.0  104.0         2          1      7
6  194294715  apartment  2100000.0  160.0         3          2      1

  furnished rent     city                region
1        no   no    cairo         new heliopolis
3        no   no  sharqia        10th of ramadan
4        no   no    cairo  new cairo - el tagamoa
5        no   no    cairo              nasr city
6        no   no    cairo         mostakbal city
```

```
!pip install pandera[io]

import pandera as pa
from pandera import Column, DataFrameSchema, Check, Index
inferred_schema = pa.infer_schema(data_cleaned).to_script()
print(inferred_schema)
```

Show hidden output

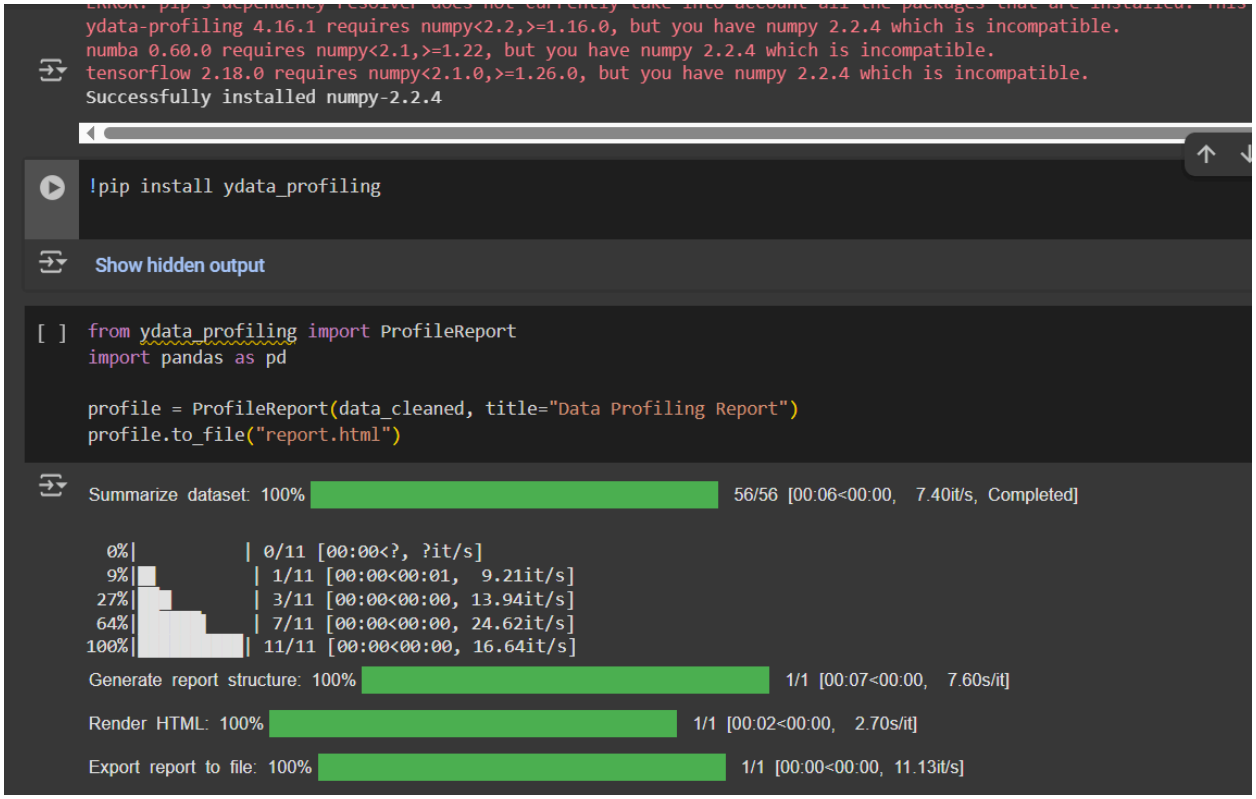
```
print(data_cleaned.dtypes)
```

## . Automated Data Profiling After Cleaning

Tool: ydata-profiling

- Profile report generated with:
    - Data types
    - Missing values summary
    - Correlation matrix
    - Duplicates
    - Descriptive statistics

📄 Saved Output: `report.html`



## Summary:

Dataset cleaned, validated, and ready for modeling or EDA.

- All fields are now with the suitable datatypes.
- Outliers managed, nulls removed, and structure validated.